

(MATH203) Applied Linear Algebra

Mathematical Concepts

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0 Notation and Conventions

General mathematical notation and writing follow *(MATH000) Mathematical Language*. The conventions below record only notation specific to linear algebra used in this book. When several equivalent notations are listed, the first form is used throughout.

Linear Algebra

Notation	Meaning
$\mathbf{A} = (a_{ij})$	matrix with (i, j) -entry a_{ij}
$\mathbf{A}^\top$	transpose of $\mathbf{A}$
$\mathbf{A}^H$	conjugate transpose of $\mathbf{A}$
$\mathbf{A}^{-1}$	inverse of $\mathbf{A}$
$\mathbf{I}_n$	$n \times n$ identity matrix
$\mathbf{0}$	zero vector or zero matrix, with dimension understood
$\mathbf{D}$	diagonal matrix when a local name is required
$\mathbf{P}$	projection matrix when a local name is required
$\text{diag}(d_1, \dots, d_n)$	diagonal matrix with entries $d_1, \dots, d_n$
$\mathbf{1}$	all-ones vector
$\det(\mathbf{A}), \mathbf{A} $	determinant
$\text{rk}(\mathbf{A}), \text{rank}(\mathbf{A})$	rank
$\text{tr}(\mathbf{A})$	trace
$\text{adj}(\mathbf{A})$	adjugate matrix
$\text{Col}(\mathbf{A})$	column space
$\text{Row}(\mathbf{A})$	row space
$\text{Null}(\mathbf{A}), \ker \mathbf{A}$	null space
$\ker T$	kernel of a linear map T
$\text{im } T$	image of a linear map T
$\mathbf{e}_i$	i th standard basis vector
$\dim V$	dimension of V
$\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$	span of the listed vectors
$U \oplus V$	direct sum
$\langle \mathbf{u}, \mathbf{v} \rangle, (\mathbf{u}, \mathbf{v})$	inner product; over complex spaces, conjugate-linear in the first argument and linear in the second
$\mathbf{u} \cdot \mathbf{v}, \mathbf{u}^\top \mathbf{v}$	standard Euclidean inner product on $\mathbb{R}^n$
$\mathbf{u}^H \mathbf{v}$	standard Hermitian inner product on $\mathbb{C}^n$
$\ \mathbf{v}\ $	norm of $\mathbf{v}$
$\mathbf{u} \perp \mathbf{v}$	orthogonality
$U^\perp$	orthogonal complement of the subspace U
$\text{proj}_U \mathbf{x}$	orthogonal projection of $\mathbf{x}$ onto the subspace U
$\text{proj}_{\mathbf{v}} \mathbf{x}$	orthogonal projection of $\mathbf{x}$ onto $\text{Span}(\mathbf{v})$
$\mathbf{u} \times \mathbf{v}$	cross product in $\mathbb{R}^3$
$p_{\mathbf{A}}(\lambda), \chi_{\mathbf{A}}(\lambda)$	characteristic polynomial of $\mathbf{A}$
$m_{\mathbf{A}}(\lambda), \mu_{\mathbf{A}}(\lambda)$	minimal polynomial of $\mathbf{A}$
λ, λ_i	eigenvalue of $\mathbf{A}$
$\mathbf{v}, \mathbf{v}_i$	eigenvector of $\mathbf{A}$
$E_\lambda(\mathbf{A}), V_\lambda(\mathbf{A})$	eigenspace of $\mathbf{A}$ associated with λ

Notation	Meaning
$\sigma(\mathbf{A}), \text{spec}(\mathbf{A})$	spectrum of $\mathbf{A}$
$\rho(\mathbf{A})$	spectral radius
$\sigma_i(\mathbf{A})$	i th singular value of $\mathbf{A}$

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